



OPERATING SYSTEM

What do you know about Operating Systems?

OPERATING SYSTEM



0:18 / 1:30 • Definição >



The operating system's job

Your computer's **operating system (OS)** manages all of the **software** and **hardware** on the computer. Most of the time, there are several different computer programs running at the same time, and they all need to access your computer's **central processing unit (CPU)**, **memory**, and **storage**. The operating system coordinates all of this to make sure each program gets what it needs.



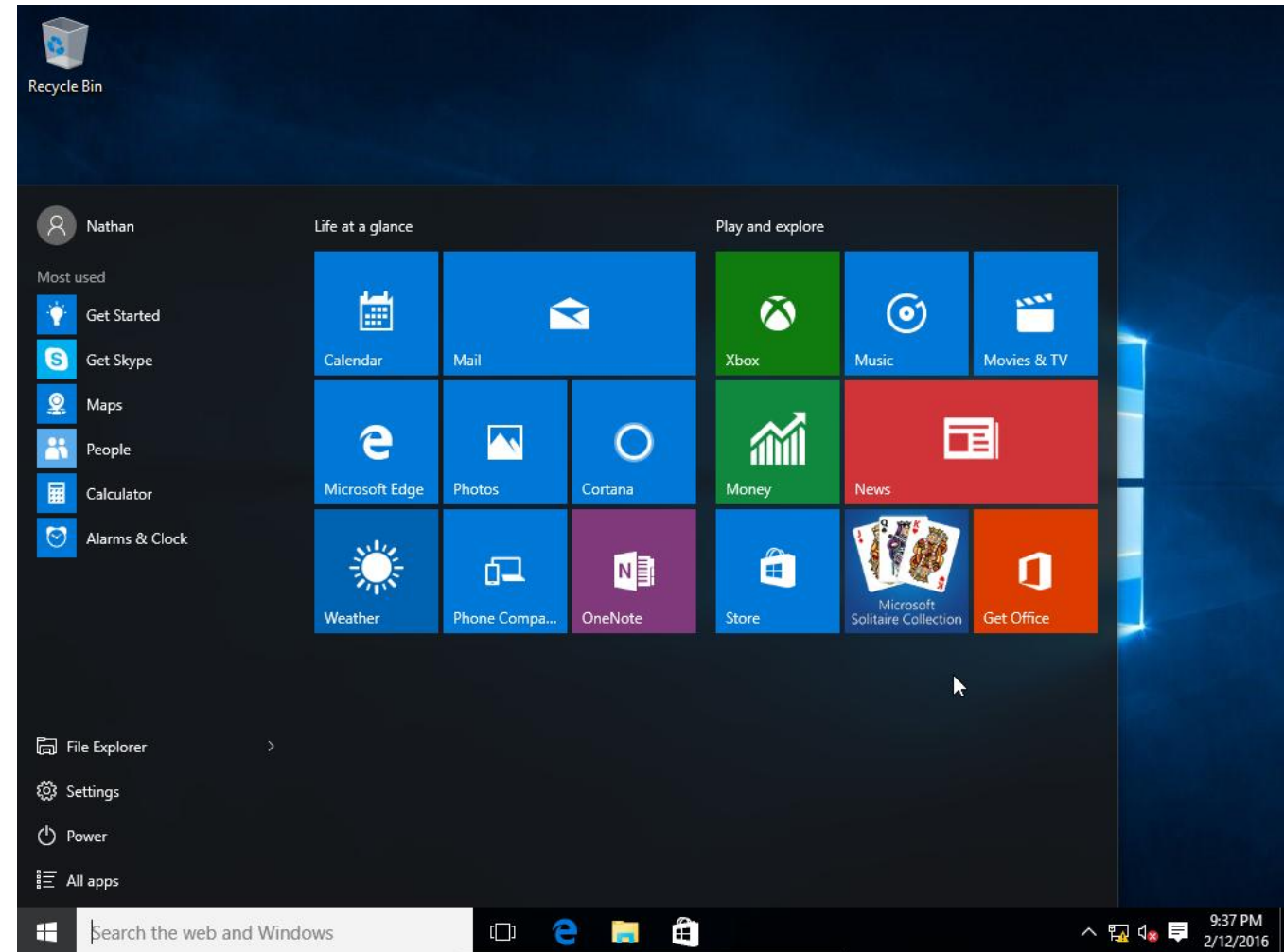
Types of operating systems

- Operating systems usually come **pre-loaded** on any computer you buy. Most people use the operating system that comes with their computer, but it's possible to upgrade or even change operating systems. The three most common operating systems for personal computers are **Microsoft Windows**, **macOS**, and **Linux**.



Types of operating systems

- Modern operating systems use a **graphical user interface**, or **GUI** (pronounced **gooey**). A GUI lets you use your mouse to click **icons**, **buttons**, and **menus**, and everything is clearly displayed on the screen using a combination of **graphics** and **text**.



Operating systems for mobile devices

- The operating systems we've been talking about so far were designed to run on **desktop** and **laptop** computers. **Mobile devices** such as **phones**, **tablet computers**, and **MP3 players** are different from desktop and laptop computers, so they run operating systems that are designed specifically for mobile devices. Examples of mobile operating systems include **Apple iOS** and **Google Android**.
- Operating systems for mobile devices generally aren't as fully featured as those made for desktop and laptop computers, and they aren't able to run all of the same software. However, you can still do a lot of things with them, like watch movies, browse the Web, manage your calendar, and play games.



Examples of Operating Systems

Microsoft Windows

- Microsoft created the **Windows** operating system in the mid-1980s. There have been many different versions of Windows, but the most recent ones are **Windows 10** (released in 2015), **Windows 8** (2012), **Windows 7** (2009), and **Windows Vista** (2007). Windows comes **pre-loaded** on most new PCs, which helps to make it the **most popular operating system** in the world.



Examples of Operating Systems

macOS

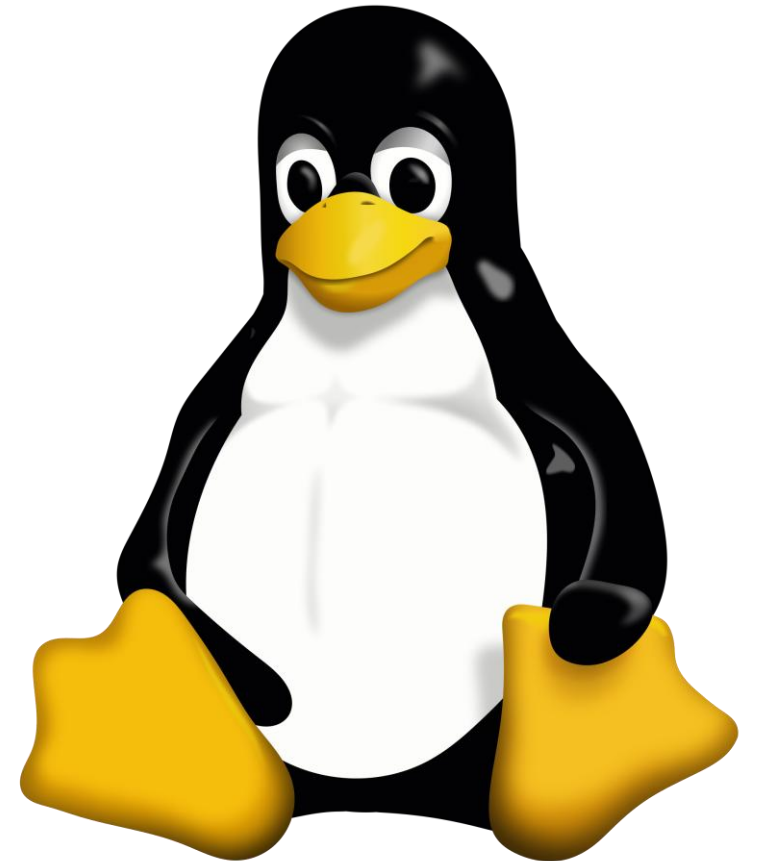
- **macOS** (previously called **OS X**) is a line of operating systems created by Apple. It comes preloaded on all Macintosh computers, or Macs. Some of the specific versions include **Catalina** (released in 2019), **Mojave** (2018), **High Sierra** (2017), and **Sierra** (2016).
- According to **StatCounter Global Stats**, macOS users account for less than **10%** of global operating systems—much lower than the percentage of Windows users (more than **80%**). One reason for this is that Apple computers tend to be more expensive. However, many people do prefer the look and feel of macOS over Windows.



Examples of Operating Systems

Linux

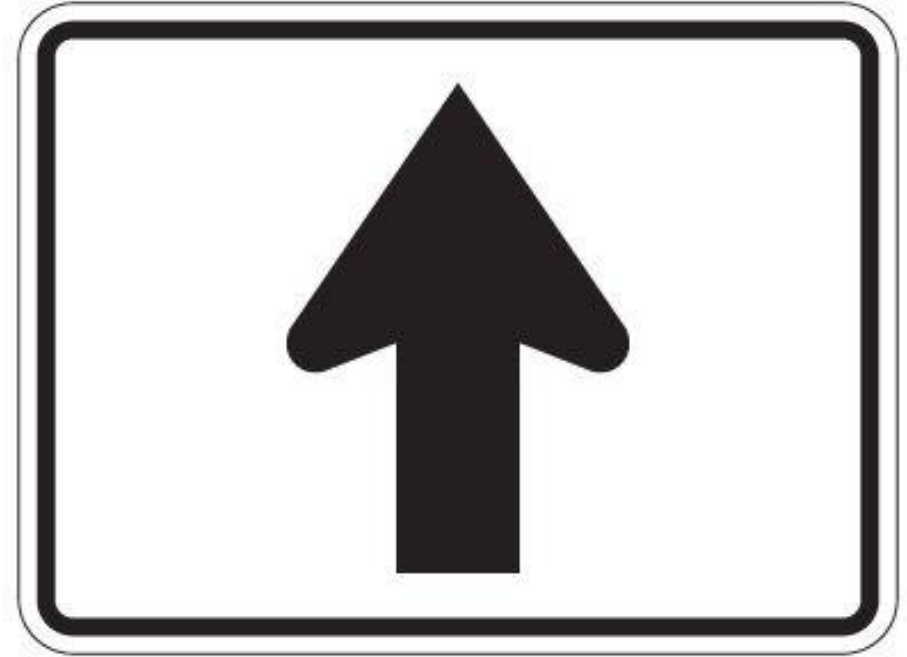
- **Linux** (pronounced **LINN-ux**) is a family of **open-source** operating systems, which means they can be modified and distributed by anyone around the world. This is different from **proprietary software** like Windows, which can only be modified by the company that owns it. The advantages of Linux are that it is **free**, and there are many different **distributions**—or versions—you can choose from.
- According to **StatCounter Global Stats**, Linux users account for less than **2%** of global operating systems. However, most **servers** run Linux because it's relatively easy to customize.



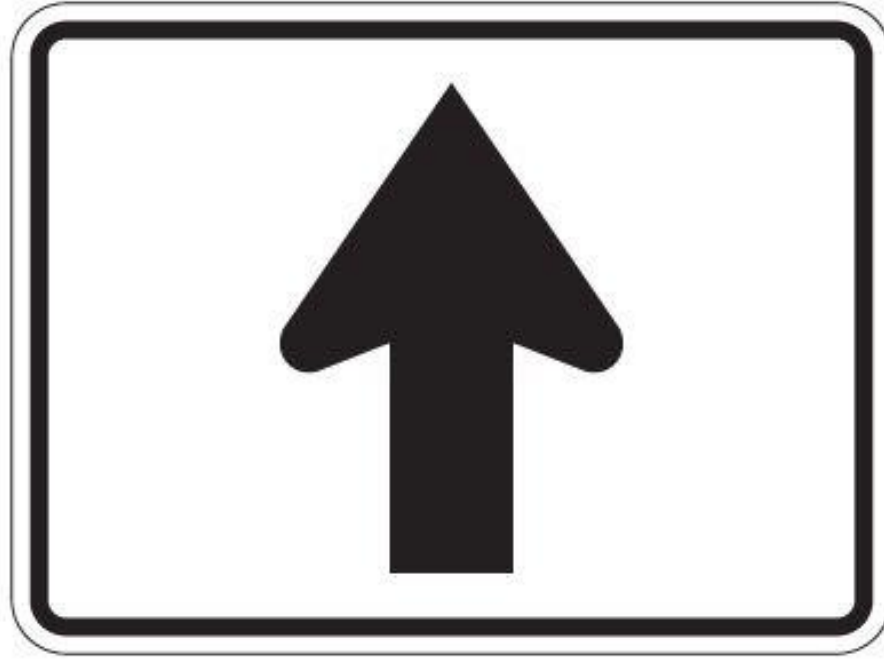
Phrasal Verbs in IT

Guess the verb by
looking at the
images

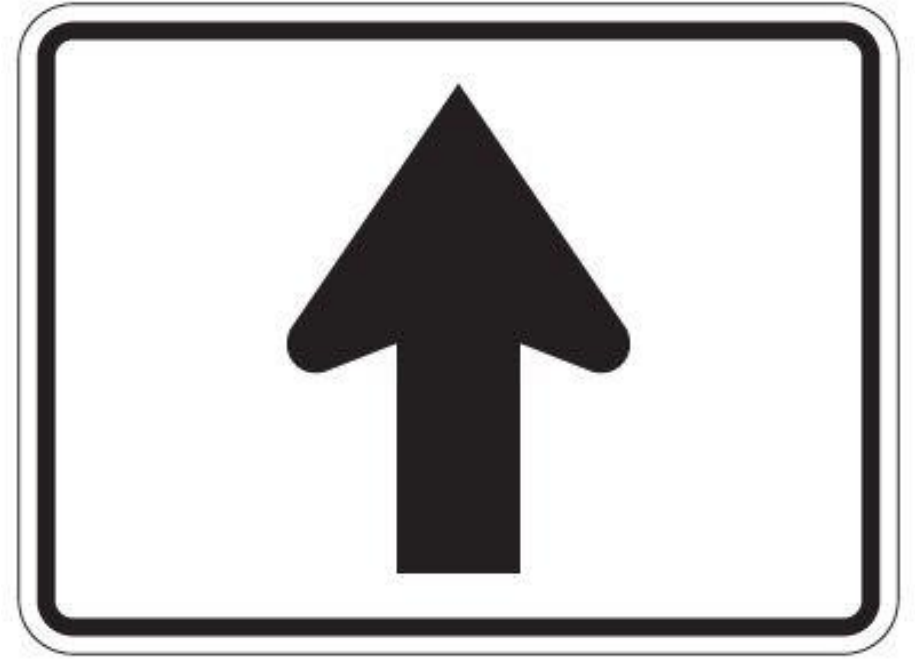
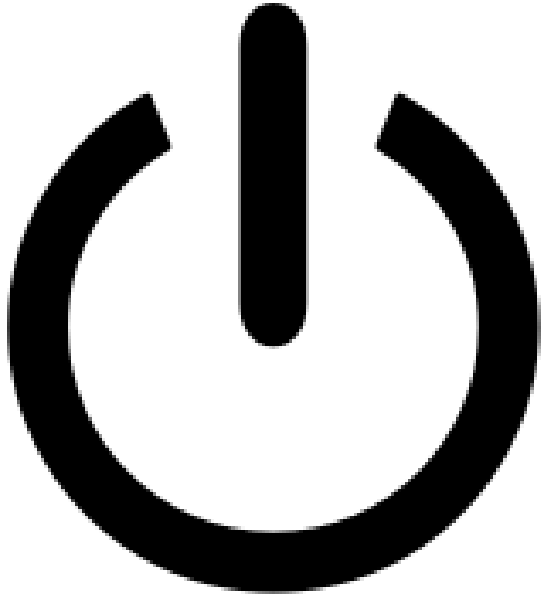




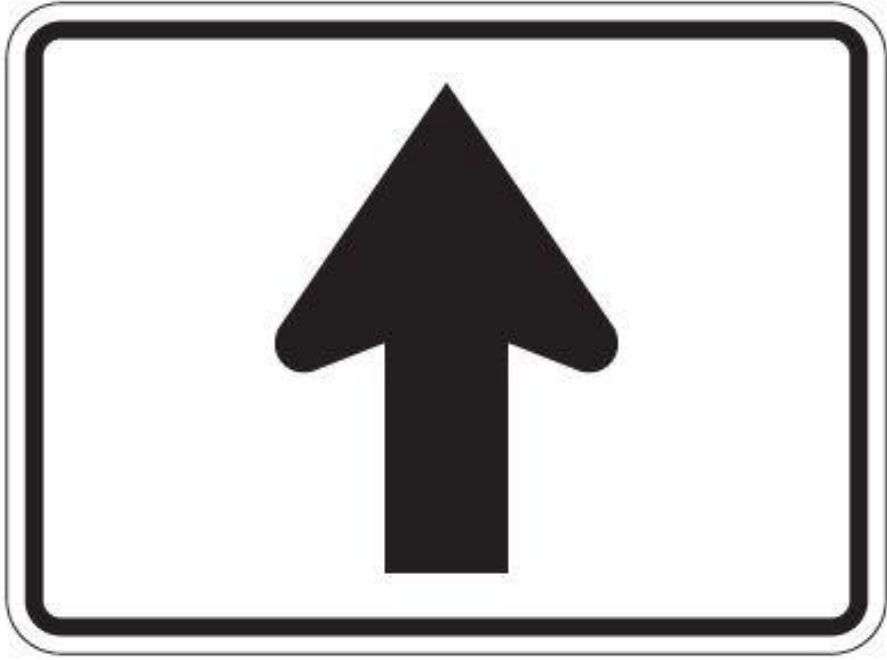
Back up: to make an extra copy of computer information



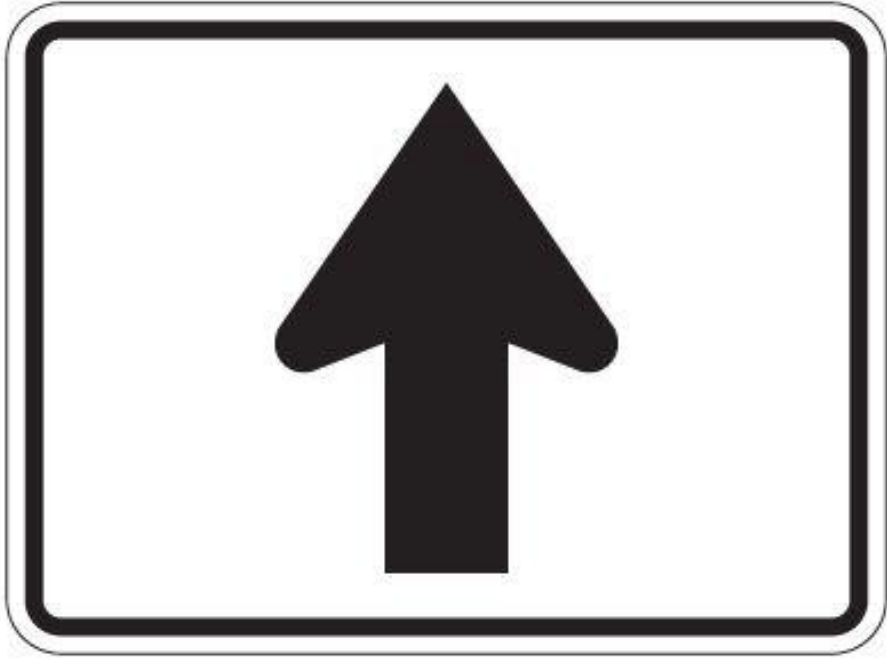
Catch up: to do something you did not have time to do earlier



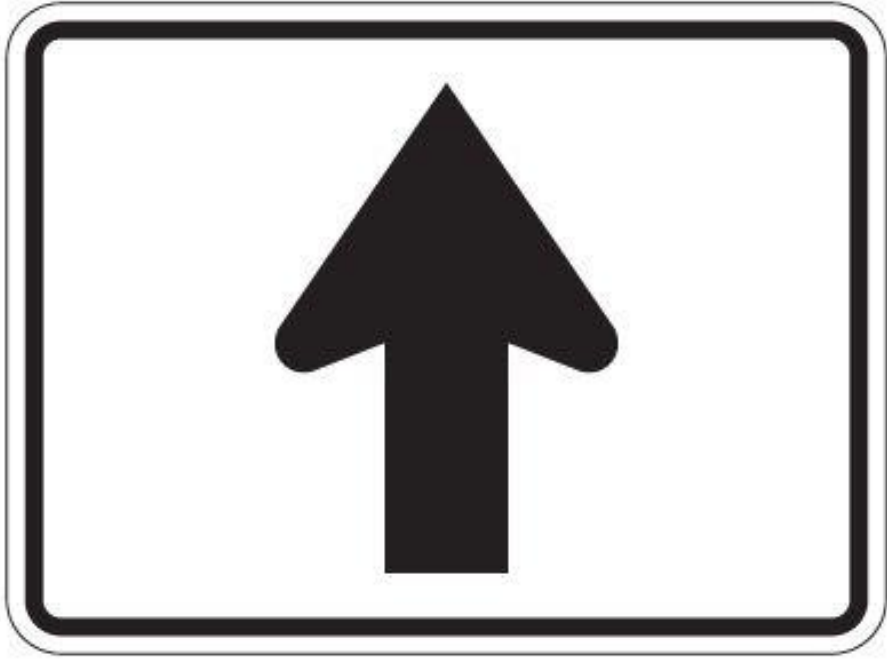
Start up: free up the process a computer goes through when it starts operating after being switched on



Update: to make something more modern or suitable for use now by adding new information or changing its design

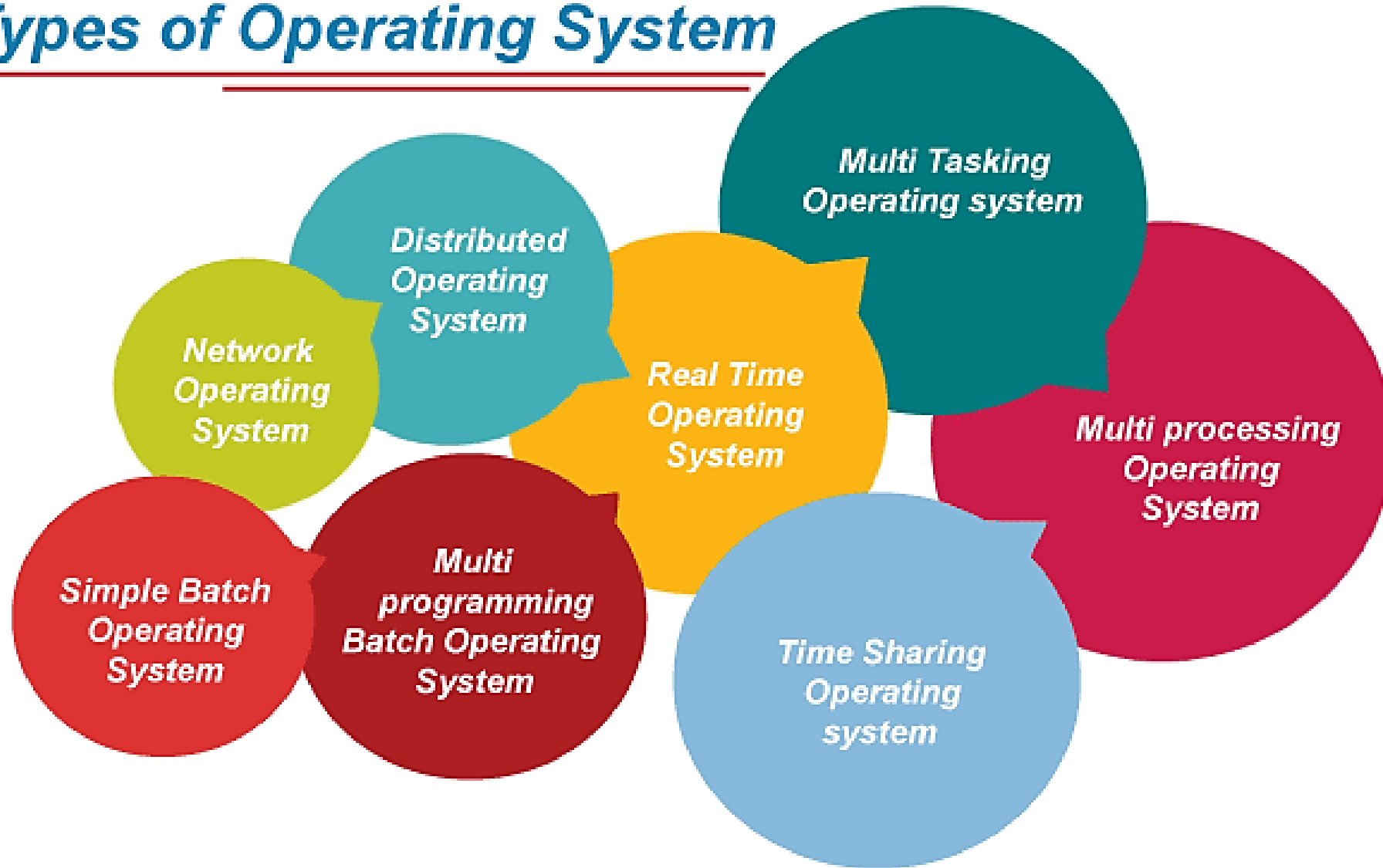


Upgrade: a piece of software or equipment that improves the quality or usefulness of a computer or machine.



Upload: to copy or move programs or information to another computer system, especially to the internet or a larger computer.

Types of Operating System



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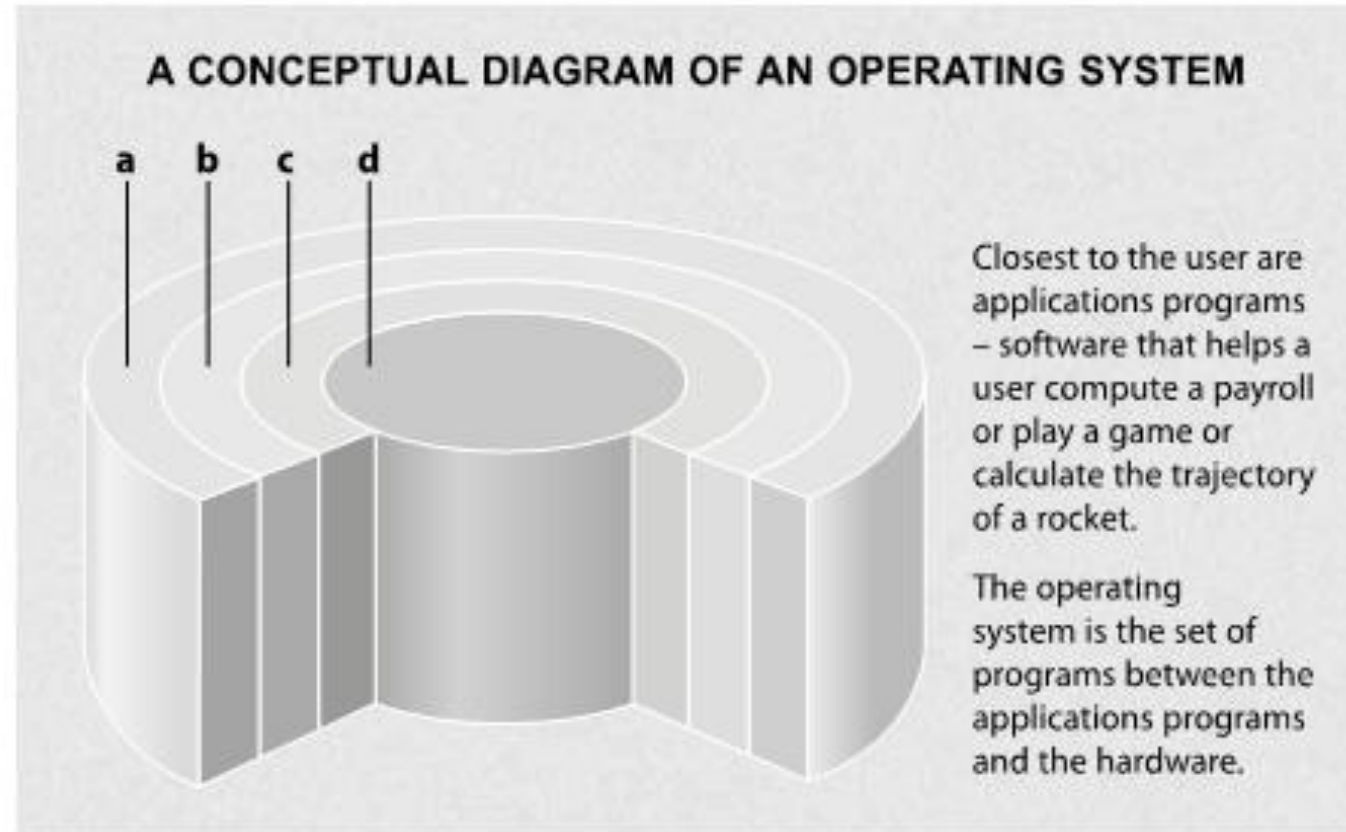
Ex. 5 - Associate the different modes of operation to the following texts.



- a) Real – time processing: is used when the computer has to react within a guaranteed time to an input. It does not necessarily have to be fast. It simply has to be quick enough to respond to inputs in a predictable way. Computers using this mode of operation are often dedicated to the control of systems such as industrial processes, planes and space flights.
- b) Multi - programming: is a method of operating in which several programs appear to be running at once. It switches jobs in and out of processor time according to priority. For example, while one job is being allocated printer time, another will be being processed in memory. The processor is so fast that it seems that many jobs are being processed at the same time.
- c) Batch processing: it is where programs or data are collected together and processed in one go. Typically, the processing of payrolls, electricity bills, invoices and daily transactions are dealt with this way. This method of operation lends itself to jobs with similar inputs, processing and outputs where **no human intervention** is needed. Jobs are stored in a queue until the computer is ready to deal with them.
- d) Interactive processing: This system is used when the tasks on the computer system require a continual exchange of information between the user and the computer system. It can be seen as the opposite of batch processing.
- e) Multi-tasking: This is not just about running more than one application at the same time. It allows various tasks to run concurrently, taking turns using the resources of the computer. This can mean running a couple of applications, sending a document to the printer and downloading a web page.
- f) Multi-access and multi-user: In this mode of operation, several users can use the same system together via a LAN. The CPU deals with users in turn; clearly the more users, the slower the response time. Generally, however, the processor is so fast that the response time at the most is a fraction of a second and the user feels they are being dealt with immediately.

6. Match the labels to the four layers of this diagram with the help of the diagram caption.

B	Applications programs
A	User
D	Hardware
C	operating system



7. Read the following text carefully and answer the questions below.

OPERATING SYSTEMS: HIDDEN SOFTWARE

When a brand-new computer comes off the factory assembly line, it can do nothing. The hardware needs software to make it work. Are we talking about applications software such as word processing or spreadsheet software? Partly. But an applications software package does not communicate directly with the hardware. Between the applications software and the hardware is a software interface – an operating system. An operating system is a set of programs that lies between applications software and the computer hardware.

The most important program in the operating system, the program that manages the operating system, is the supervisor program, most of which remains in memory and is thus referred to as resident. The supervisor controls the entire operating system and loads into memory other operating system programs (called non-resident) from disk storage only as needed.

An operating system has three main functions: 1) manage the computer's resources, such as the central processing unit, memory, disk drives, and printers, 2) establish a user interface, and 3) execute and provide services for applications software. Keep in mind, however, that much of the work of an operating system is hidden from the user. In particular, the first listed function, managing the computer's resources, is taken care of without the user being aware of the details. Furthermore, all input and output operations, although invoked by an applications program, are actually carried out by the operating system.

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7.1. Find synonyms for the underlined words.

but – however

however – nevertheless

thus – consequently

furthermore – moreover

such as – for instance

although – despite (+ing)

7.2. Try to find the answers to the following questions.

- a) What difference is there between applications software and operating systems?

Applications software does not communicate directly with computer hardware.

- b) Why is the supervisor program the most important operating system program?

Because it controls the entire OS and loads the other OS programs into memory as needed.

- c) What is the difference between resident and non-resident programs?

Programs, which remain in memory, are resident.

Programs, which are loaded from disk storage, are non-resident.

- d) What are the main functions of an operating system?

To manage the computer resources, to establish user interface and to execute and provide services for app software.

The background of the image is a dense, chaotic pile of three-dimensional white letters and numbers. These characters are scattered across the entire frame, creating a textured, almost abstract surface. The lighting is soft and even, highlighting the edges of the characters. In the center of this sea of characters, the words "Linking words" are written in a clean, yellow, sans-serif font. The text is centered both horizontally and vertically, standing out clearly against the busy, monochromatic background.

Linking words

Linking words



- Cohesive devices that are used to connect ideas
- They can be in different positions in a sentence

***MOREOVER**, there is a growing reliance on these networks to access information and services.*

*There is, **MOREOVER**, a growing reliance on these networks to access information and services.*

- Connect two sentences to make a longer sentence
- Their main broad types are: result, examples, time, addition, contrast, comparison

Operating Systems

Linking words

Adding another point

Concluding

Announcing a change of subject

Sequencing points

Emphasizing a contrast

Giving examples

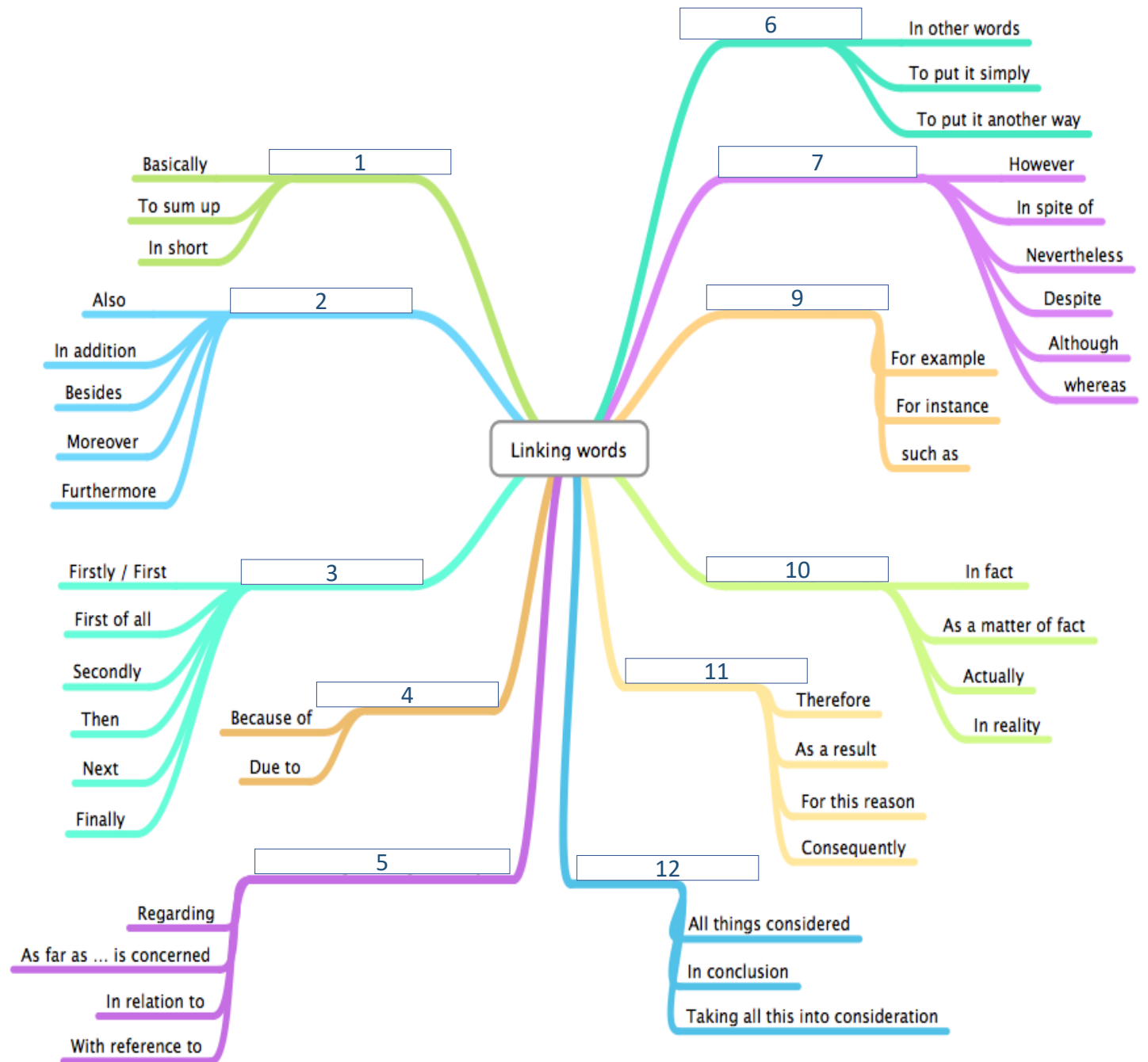
Expressing reality

Explaining

Giving a cause

Summarizing

Giving a result



7.3. Complete the gaps in this summary of the text on operating systems using these linking words and phrases:

although

because

but

in addition

such as

therefore

The user is aware of the effects of different applications programs ¹⁾ but operating systems are invisible to most users. They lie between applications programs, ²⁾ such as word processing, and the hardware. The supervisor program is the most important. It remains in memory, ³⁾ therefore it is referred to as resident. Others are called non-resident ⁴⁾ because they are loaded into memory only when needed. Operating systems manage the computer's resources, ⁵⁾ such as the central processing unit. ⁶⁾ In addition, they establish a user interface, and execute and provide services for applications software. ⁷⁾ Although input and output operations are invoked by applications programs, they are carried out by the operating system.